UNIT II: Linear Data Structures – Stacks and Queues

Array Representation of Stacks – Operations on a Stack – Applications of Stacks – Infix to Postfix conversion — Evaluating Postfix expressions – Queues – Array Representation of Queues – Linked Representation of Queues – Circular Queues – DeQueue – Applications of Queues.

STACK

What is a Stack?

A **stack** is a linear data structure.

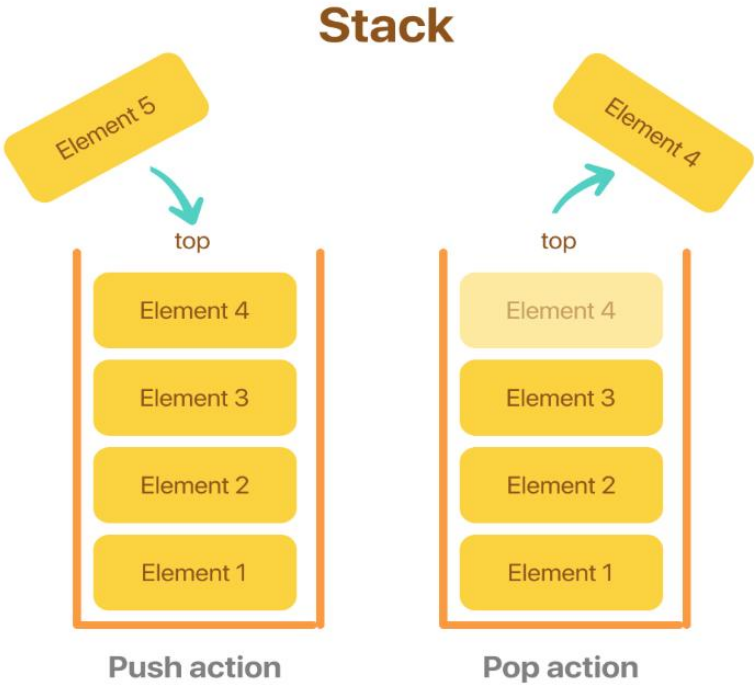
"A **Stack** follows the **LIFO** principle — *Last In, First Out*.

It works like a **pile of books** — you can only add or remove the topmost book.

Think of it like a stack of plates in a cafeteria. You always remove the top plate first.

Operation	Meaning
Push(x)	Insert x at the top of stack
Pop()	Remove and return top element
Peek()	Return top element without removing
isEmpty()	Returns true if stack is empty
isFull()	Returns true if stack is full

Time complexity is O(1) for all these operations."



ARRAY REPRESENTATION OF STACK

```
#define SIZE 100
int stack[SIZE];
int top = -1;  // Stack is empty

// Push operation
void push(int value) {
    if (top >= SIZE - 1)
        printf("Stack Overflow\n");
    else
        stack[++top] = value;
}

// Pop operation
int pop() {
    if (top < 0)
        printf("Stack Underflow\n");
    else
        return stack[top--];
}
```

Definition: Array Implementation of Stack

1. **Array implementation of a stack** refers to using a fixed-size array to store elements of a **stack** data structure, where elements follow the **LIFO** (Last In, First Out) principle.

2. The operations like **push**, **pop**, and **peek** are performed using an integer variable called top, which keeps track of the index of the top element in the array.

Key Concepts:

- **Stack:** A linear data structure where the last inserted element is the first to be removed.
- **Array:** A contiguous memory block used to hold the stack elements.
- **Top:** An index pointing to the most recently inserted element.

Algorithm for PUSH (Insert Element)

Input: Stack array stack[], top pointer top, maximum size MAX, element item to insert
Output: Updated stack with item added at the top

```
Algorithm PUSH(stack, top, MAX, item)
1. if top == MAX - 1 then
2.     print "Stack Overflow"
3.     return
4. else
5.     top ← top + 1
6.     stack[top] ← item
7.     print "Item inserted"
8. end if
```